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PALEOMAGNETISM OF LATE PALEOZOIC ROCKS IN THE TIANSHAN, NORTHWESTERN CHINA

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Abstract. A paleomagnetic study of late Paleozoic rocks in two areas of northwestern China helps to constrain the timing of collision between Tarim and the Junggar part of Eurasia along Tianshan. A primary Late Permian direction (D/I = 198/-59, K=24, A95=12.6) is derived from seven sites (53 samples) from the Heavenly Lake (Tianzhi) section in northern Tianshan that has both normal and reversed polarities and passes a fold test at the 99% confidence level. This result gives a latitude of 39°N in the Late Permian for southern Junggar. Twenty-two sites (144 samples) from the Devonian and Carboniferous rocks across Tianshan reveal postfolding remagnetizations some of which are interpreted as being acquired in the Permian due to the Tarim-Eurasia collision. The Late Permian paleomagnetic pole from the Heavenly Lake section is statistically indistinguishable from three published Late Permian poles from Tarim indicating closure of the Tianshan ocean and suturing of Tarim and Junggar before that time. These paleomagnetic poles from northwestern China are, however, significantly different from those of stable Eurasia, suggesting substantial later relative motion that can be partly accounted for by the progressive closure of the Mongolo-Okhotsk ocean and associated tectonic rotations. The absence of a discernable Permian ocean record west of the Khangay highland in central Mongolia requires that the rotations be accommodated within a diffusive plate boundary in the Eurasian continent. A Late Permian continental plate reconstruction is presented that takes into consideration these new paleomagnetic results and other geological constraints.

INTRODUCTION

China comprises at least 11 different continental fragments and arc-related assemblages, including three major continental blocks of South China (equivalent to Yangtze), North China (equivalent to Sino-Korean), and Tarim, and eight smaller continental and arc-related fragments, with Junggar in the northwest, Dahinggan and Xingkai-Bureya in the northeast, and Qaidam (QD), Central Kunlun (CKL), Eastern Qiangtang (EQ), Western Qiangtang (WQ), and Lhasa in the southwest (Figure 1). Biogeographically, Junggar, Dahinggan, and Xingkai-Bureya in the north are closely related to the Siberia (=Angaran Shield) [Rong and Zhang, 1982; Li and Yao, 1983; Yin, 1989; Ziegler, 1990]. Western Qiangtang and Lhasa in the south clearly have Gondwanan affinities, judging from glaciogenic deposits of late Paleozoic age [Liang et al., 1984; Lin et al., 1989] and faunal characteristics [Jin, 1985]. The remaining continental blocks of South China, North China, Tarim, Eastern Qiangtang, and arc-related fragments of Qaidam and Central Kunlun are believed to have resided within the late Paleozoic Tethys [Sengör, 1987; Nie et al., 1990], though in the early Paleozoic at least one of them (i.e., South China) was most likely part of Eastern Gondwana from which it subsequently rifted in the Devonian [Nie and Rowley, 1991]. One of the keys to understand tectonic evolution of Asia is the timing of continental collision between these once separated blocks. In a general sense, there appears to be a trend that from the north to the south, the time of suturing becomes younger and younger, reflecting progressive addition of micro-continents onto the Eurasia from the latest Paleozoic to the Eocene (for details, see, for example, Zhang et al. [1984] and Sengör [1987]), with one important exception being represented by the Mongolo-Okhotsk ocean which did not close until the Late Jurassic to Early Cretaceous [Kosygin and Parfenov, 1981; Rowley et al., 1985; Nie et al., 1990; Zonenshain et al., 1990].

Tarim, Kazakhstan, and the Altaide margin of Siberia are juxtaposed along the Tianshan (=Tien Shan of Soviet usage) system in Xinjiang province of northwestern China. Tianshan forms the geographic boundary between the Junggar basin to the north and the Tarim basin to the south, and within it are the Turpan basin in the east and Yining basin in the west (Figure 2). The NW-SE trending Irtysh-Bayatik (IT-BY) suture separates Siberia from Kazakhstan prior to its Late Carboniferous closure (Figure 1) [Rowley et al., 1985; Zonenshain et al., 1990] and clearly extends into China through northwestern Junggar. This boundary extends southeastward from the ophiolitic melanges north of Karamay to the Kelameili ophiolites within the southern Altay in northeastern Junggar. This southeast trend is almost perpendicular to the present structural trend of northwestern Junggar, which reflects more recent deformation. Most of Junggar between this Irtysh-Bayatik and Kelameili line in the north and northern Tianshan in the south was apparently the eastern extension of Kazakhstan. Junggar, along with northern Tianshan, therefore represents the part of the late Paleozoic continental margin of the already amalgamated Siberia and Kazakhstan blocks.

Collision along Tianshan between Tarim and Junggar has been inferred to be in the Early Permian based on geological features such as termination of andesitic volcanism, the development of unconformities between late Early Permian and underlying units [Xinjiang Stratigraphic Group, 1981], the intrusion of widespread syncollisional to postcollisional granites in Tianshan [Xinjiang Bureau of Geology and Mineral Resources, 1985; Coleman, 1989], and mixture of Angaran and Cathaysian flora across Tianshan [Wu, 1983; Hu, 1985]. This interpretation has recently been challenged by McFadden et al. [1988], who proposed that there was still a wide ocean between Tarim and Kazakhstan in the Late Permian to Early Triassic, based on a comparison of paleomagnetic results. We summarize our paleomagnetic study in Tianshan that provides new constraint on both the timing of collision between Tarim and Eurasia and also the paleolatitudinal position of northern Tianshan/southern Junggar. Our samples were collected in 1987 and 1988 from the Upper Permian rocks from the Heavenly Lake section in northern Tianshan, and from the Carboniferous and Devonian rocks from the trans-Tianshan Dushanzhi-Kuqa highway, which is still not open to foreign travel (Figure 2).

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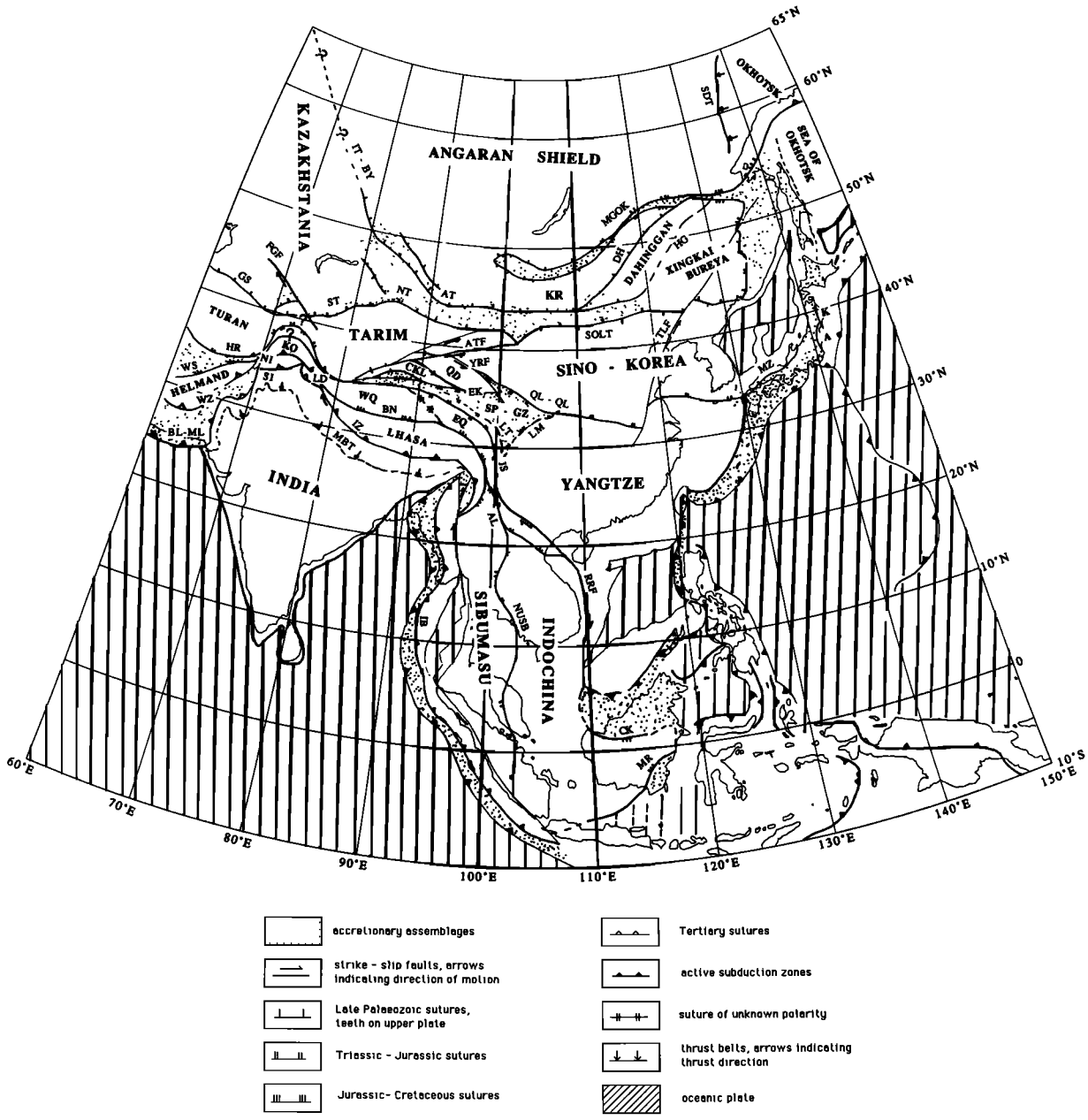


Fig. 1. Tectonic subdivisions and sutures for the eastern Asian blocks [after Nie et al. 1990]. Microcontinents or arc-related fragments: A, Abukuma; CKL, Central Kunlun; DH, Dahinggan; EQ, Eastern Qiangtang; K, Kitikami; KO, Kohistan; QD, Qaidam; and WQ, Western Qiangtang. Sutures: AL, Ailaoshan; AT, Altai; BL-ML, Bela-Muslimbagh; BN, Banggong Co-Nujiang; CK, Crocker; DH, Dahinggan; EK, East Kunlun; GS, Ghissar; HG, Heganshan; HR, Herat; IB, Indo-Burma; IT-BY, Irtysh-Bayitik; IZ, Indus-Zangbo; JS, Jinshajiang; KR, Kerulen; LD, Ladakh; LM, Longmenshan; LT, Litang; MGOK, Mongolo-Okhotsk; MR, Meratus; MZ, Maizuru; NI, North Indus; NT, North Tianshan; NUSB, Nan-Uttaradit-Sra-Kaeo-Bentong; QL-QL, Qilian-Qinling; SI, South Indus; SOLT, Solon Obo-Linxi-Tumen; ST, South Tianshan; WS, Waser; and WZ, Waziristan. Major strike-slip faults: ATF, Altyn Tagh; FGF, Fergana; RRF, Red River; TLF, Tan-Lu; and YRF, Yellow River. Thrust belts: MBT, Main Boundary Thrust; and SDT, Sette-Daban Thrust. The accretionary complex is SP-GZ, Songpan-Ganzi.

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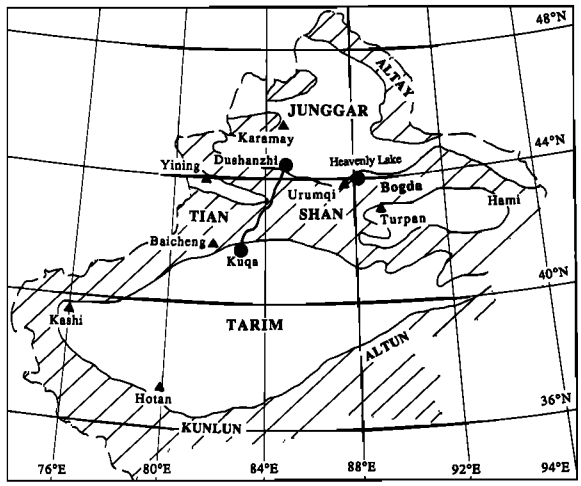


Fig. 2. Locations of study area and major geographic features of Xinjiang Province. Large solid circles give locations of the Heavenly Lake, Dushanzhi, and Kuqa; other cities and localities discussed are indicated by the triangles. Dotted line represents the provincial boundary, while dash-dot line marks the national boundary. Diagonally ruled patterns represent mountainous areas, whereas the blank spaces within Xinjiang indicate the extent of the four present-day basins.

LABORATORY PROCEDURES

Laboratory measurements were taken using a cryogenic magnetometer in the paleomagnetic laboratory at the University of Michigan that is equipped with a shielded room having a magnetic field of less than 200 nT. Thermal demagnetization proved to be most effective for separating different magnetic components in the samples and was subsequently the only demagnetization method applied. One specimen per site was chosen to perform a pilot-trial study to analyze the general demagnetization behavior of the samples. During the pilot-trial, the samples were heated at intervals of 50°C below 350°C, and 20°C or 30°C above 350°C, commonly including 12 or more steps. This provided good control on the critical transitional points for demagnetization curves so that fewer steps (usually 8-10) were necessary for thermally demagnetizing the remaining samples. The NRM intensities are typically of the order of 1 mA/m for limestones and 10 mA/m or larger for clastic and volcanic rocks. Vector analysis was done with the help of Zijderveld diagrams [Zijderveld, 1967] and stereographic projection. The characteristic directions were determined using principle component analysis as developed by Kirschvink [1980].

RESULTS FROM THE HEAVENLY LAKE (TIANZHI) SECTION

Geological Setting and Sampling

Heavenly Lake (44°N, 88.1°E), a well-known tourist attraction, is located about 100 km northeast of Urumqi within the Bogda Shan, the northern range of the Tianshan system (Figure 2). Our sampling sites are located along the main road down slope beside the Shangong He (= River) to the north of the lake. Late Cenozoic to recent tectonism exposed Lower Jurassic, Upper Permian, and Carboniferous rocks in a series of thrust-related folds in our study area. The Carboniferous units include slightly metamorphosed volcanic and turbiditic sequences. The Upper Permian consists of two groups, the Upper Jijicao below and the Lower Cangfanggou above, with a regional unconformity in between [Xinjiang Stratigraphic Group, 1981, also unpublished text for 1:200,000 mapping of Urumqi area]. Only the former is exposed along this road and includes in ascending order, as illustrated in Figure 3, the Wulapo, Jinjinzhigou, Lucaogou, and Hongyanchi formations, with many horizons of bivalve, ostracod, fish, and plant fossils that provide reliable age control. The Wulapo Formation is more than 1300 m thick and dominated by feldspar-rich sandstone and siltstone, suggesting relatively short transport and fast sedimentation rates. The Jinjinzhigou Formation is 576 m thick and is dominated by gray tuffaceous clastics. The Lucaogou Formation has a thickness of 660 m and is represented by dark gray to black phosphorite-bearing sandstone and sandy shale in the lower part, and black kerogen-rich shale and minor amounts of dolomitic in the upper part. The Hongyanchi Formation, the youngest of the four, is 733 m thick and includes gray to green siltstone, shale, and sandstone in addition to a few limestone interbeds. Overall, the sedimentary facies has been interpreted as coastal marine and deltaic for the lower part of the Upper Permian and stagnant lagoonal to lacustrine for the upper part [Zhang and Wu, 1985]. The sedimentary characteristics suggest that this sequence was deposited in a foreland basin accompanied by volcanic eruption and active tectonism. The Lower Jurassic is represented by over 700 m of gray sandstone and gray to brown sandy mudstone, with many coal beds in between. Regionally, the Triassic is known to have a limited record.

Seven sites with a total of 65 samples were collected from the Upper Permian rocks in the Heavenly Lake section (Table 1). Five sites (42 to 46) were collected from a series of steeply north dipping to overturned sandstone beds within a black shale dominated sequence, belonging to the Lucaogou Formation. Graded bedding and bottom structures including flute casts observed in the field were critical for correctly establishing bedding order. Two more sites (49 and 50) were collected from a sequence of massive greenish sandstones and siltstones that are about 800 m higher in the section farther to the north. This rock unit is a part of the Hongyanchi Formation that dips to the north at about 40° and represents a part of a hanging wall anticline. Our sites represent two units

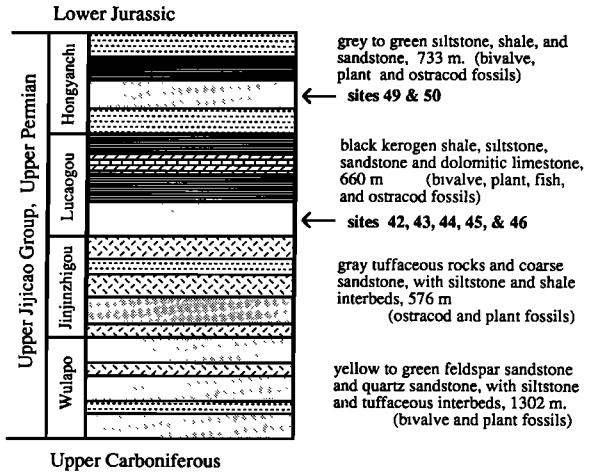


Fig. 3. Stratigraphic section for the Upper Permian section of the study area. Data are from Xinjiang Stratigraphic Group [1981]. Thicknesses are not drawn to scale but are indicated numerically. Note the stratigraphic positions of the sampling sites.

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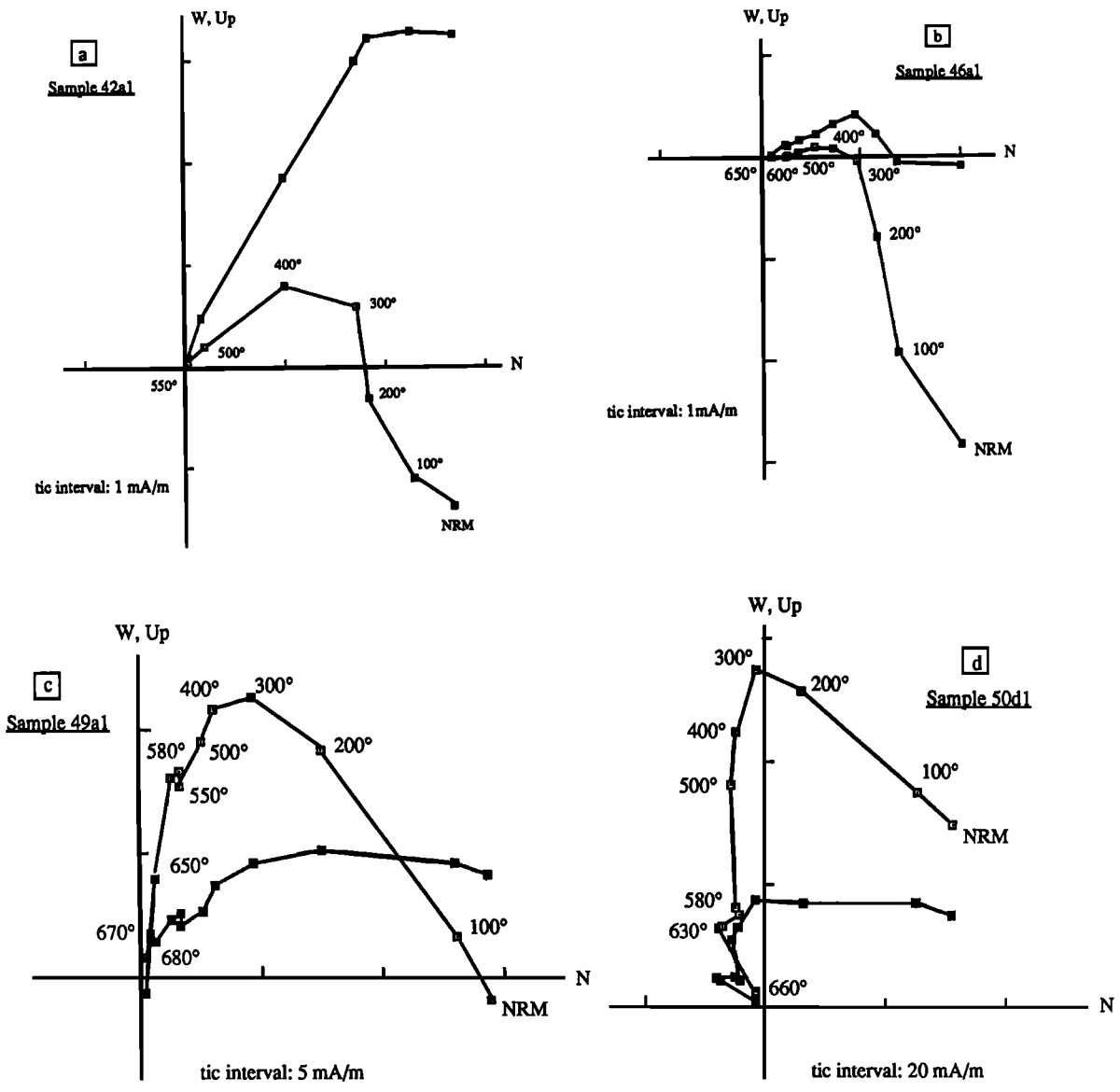


Fig. 4. Representative Zijderveld diagrams from the Heavenly Lake section. All are shown in the in situ coordinates. Open squares represent vector projection on vertical plane, solid ones horizontal plane. Figures 4a and 4b are for the Lucaogou Formation; Figures 4c and 4d are for the Hongyanchi Formation. Other details are listed in Table 1.

that have very different bedding orientations, allowing us to perform a fold test.

Summary of Paleomagnetic Directions From the Heavenly Lake

Figure 4 gives four representative demagnetization curves for the Upper Permian rocks from the Heavenly Lake section, two for the Lucaogou Formation and two for the Hongyanchi Formation. Typically, a low-temperature component is removed below 350°C that is northerly and downward with an inclination of about 60°, which is obviously a recent overprint. Another component is isolated at higher temperature ranges with blocking temperatures of about 590° and 680°, respectively. This component decays univectorially toward the origin and is distinctively different from the present-day geomagnetic field. We interpret it as the characteristic direction. For samples from sites 42, 43, 44, and 45, the high temperature component remains stable below about 590°C but becomes random afterwards, suggesting that the magnetization is mainly carried by magnetite. For samples from sites 46, 49, and 50, on the other hand, the high-temperature component remains stable up to 680°C, implying that hematite is the main magnetic carrier. Nine samples from site 46 yield stable demagnetization curves, and among them, as shown in the tilt-corrected coordinates in Figure 5, three samples give downward and northerly directions while the other six give upward and southerly directions. We interpret them as representing antipodal directions with a site mean $\alpha 95$ of 12.5. The samples with the normal and reversed polarities are from

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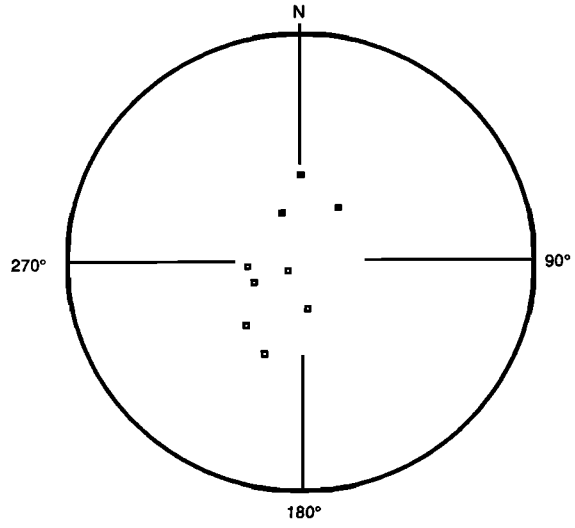


Fig. 5. Tilt-corrected directions for site 46 from the Heavenly Lake section. Solid squares are for positive inclinations and open squares for negative inclinations, indicating presence of dual polarity with a site mean α95 = 12.5.

different stratigraphic horizons that have apparently recorded frequent reversals.

Table 1 and Figure 6 summarize the results from the Heavenly Lake section. Clearly, samples from different limbs have significantly different in situ directions. As shown in Figure 6a, sites 42 to 46 give shallow and upward northwesterly directions, while sites 49 and 50 give steep and upward directions. Upon tilt correction, a much better clustering is achieved, giving a moderately steep and southwesterly direction (Figure 6b). The mean direction for all seven sites before tilt correction is Db/Ib = 307/-26, with K1=5.2 and A95=29.4, and for after tilt correction, Da/Ia =198/-59, with K2=24.0 and A95=12.6. The K2/K1 ratio of 4.6 indicates a positive fold test significant at the 99% confidence level [McElhinny, 1964]. In summary, we believe that the characteristic direction from these seven sites is a primary Late Permian direction because (1) The in situ directions are distinctively different from the present day field and the Mesozoic directions published to date from northwestern China [Li et al., 1988b; Chen et al., 1990]; (2) the directions pass a fold test; (3) site 46 records both normal and reversed directions; and (4) all samples show stable and definable demagnetization behavior. A paleomagnetic pole calculated from the tilt-corrected direction is at 76°N and 193°E, which gives a paleolatitude of 39°N for the study area (44°N, 88.1°E) using a reversed polarity option.

One point to be made is how the mixed polarity we found in this study relates to the late Paleozoic Kiaman Reversed Superchron and the overlying Illawarra Mixed Interval [Irving and Parry, 1963]. According to the latest time scale [Harland et al., 1990], the upper boundary of the Kiaman is placed at 248 Ma (p.167). Ages of 250 Ma - 247.5 Ma and 247.5 Ma - 245 Ma are now assigned to the Longtanian and Changxingian stages of South China, newly selected as the international stages for the uppermost Permian because of their abundant marine fossils [Harland et al., 1990]. The Upper Jijicao Group, from which our samples are collected, correlates with the Longtanian according to Yang et al. [1986, p.123]. Therefore, the fact that mixed polarities are recorded in site 46 is consistent with extending the Illawarra mixed interval into the mid-Tatarian or the Longtanian, in agreement with other recent findings in Tarim [McFadden et al., 1988], South China [Zhao and Coe, 1987; Heller et al., 1989; Steiner et al., 1989], and North America [Molina-Garza et al., 1989].

SUMMARY OF RESULTS FROM TIANSHAN

Geological Setting and Sampling

A total of 243 samples were collected from 43 sites along the Dushanzhi (44.3°N/84.9°E) - Kuqa (41.6°N/83°E) highway across Tianshan (Figure 2). Among them 141 samples from 22 sites give definable characteristic directions (Table 2). The remaining samples are either too weakly magnetized or yield Zijderveld diagrams from which it is impossible to infer a well defined vector. The useful samples include 78 samples (12 sites) from the Middle Carboniferous Qianxia Formation and the Late Carboniferous Shirenzhigou Formation of northern Tianshan. Both formations are clastic-dominated sequences with widely variant strikes and generally steep dips (Table 2). Four sites with a total of 27 samples were collected from the Early Carboniferous rocks in central Tianshan, including three sites (16 samples) from the light-colored Yiayungou limestone and one site (site 14, 11 samples) from the Dahajunshan rhyolites. Six sites (53 samples) are from the Middle Devonian

TABLE 1. Upper Permian Paleomagnetic Directions From Heavenly Lake (44°N, 88.1°E)

Site	Strike	Dip	n (n')	D_b	I_b	D_a	I_a	K	α_{95}	Range (°C)
42	261	106N*	8 (9)	301	-9	240	-35	161	4.4	350-550
43	237	103N	9 (10)	308	-8	190	-62	100	5.2	300-570
44	250	111N	8 (10)	328	-13	180	-54	27	11.9	300-550
45	232	102N	6 (7)	298	-9	192	-58	22	4.6	300-570
46+	240	103N	9 (14)	308	-2	208	-64	18	12.5	400-650
49	250	38N	5 (6)	301	-76	179	-62	115	7.2	400-680
50	250	38N	8 (9)	265	-79	178	-54	105	5.4	300-650

Overall: N (site) = 7, Db / Ib = 307 / -26 K_1 = 5.2 A_{95} = 29.4
n (samples) = 53, Da / Ia = 198 / -59 K_2 = 24.0 A_{95} = 12.6

Pole position after tilt correction: 76°N, 193°E, K_2 / K_1 = 4.6. Paleolatitude: 39°N

Number of samples used (sampled) is n(n'). Declination and Inclination before (after) tilt correction is Db, Ib (Da, Ia).
* Dip angle > 90° indicates overturned bed.
+ Site 46 has both normal and reversed polarities.

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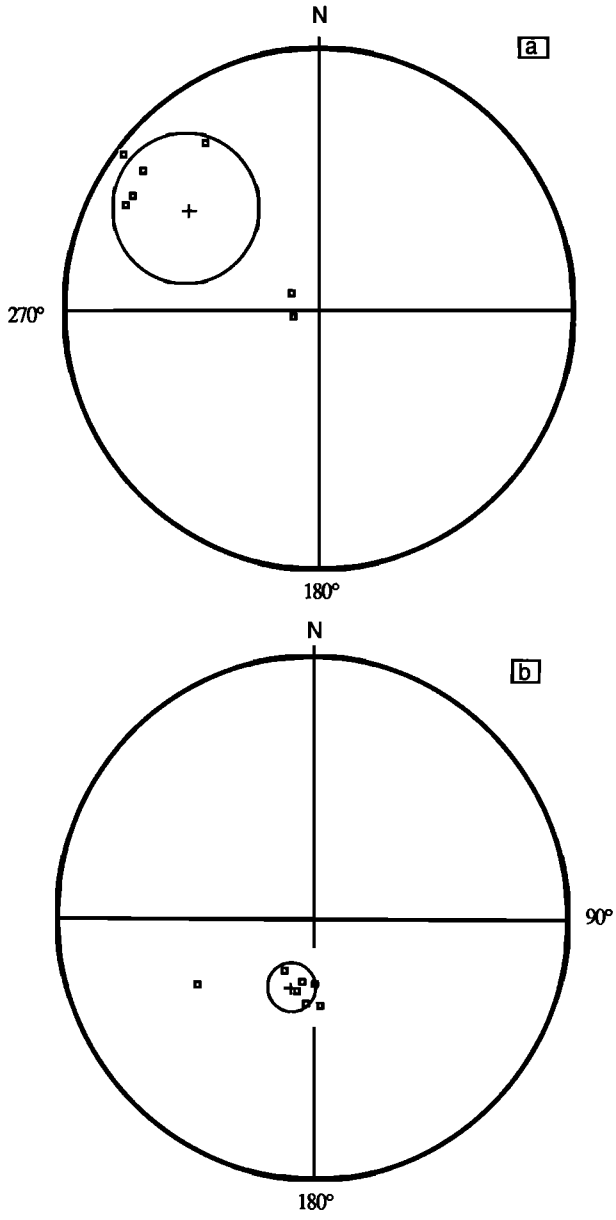


Fig. 6. (a) in situ and (b) tilt-corrected site mean directions from the Heavenly Lake section. All inclinations are negative. The 95% confidence circles are shown along with the mean values as marked by the crosses, showing a positive fold test; For details, see text and Table 1.

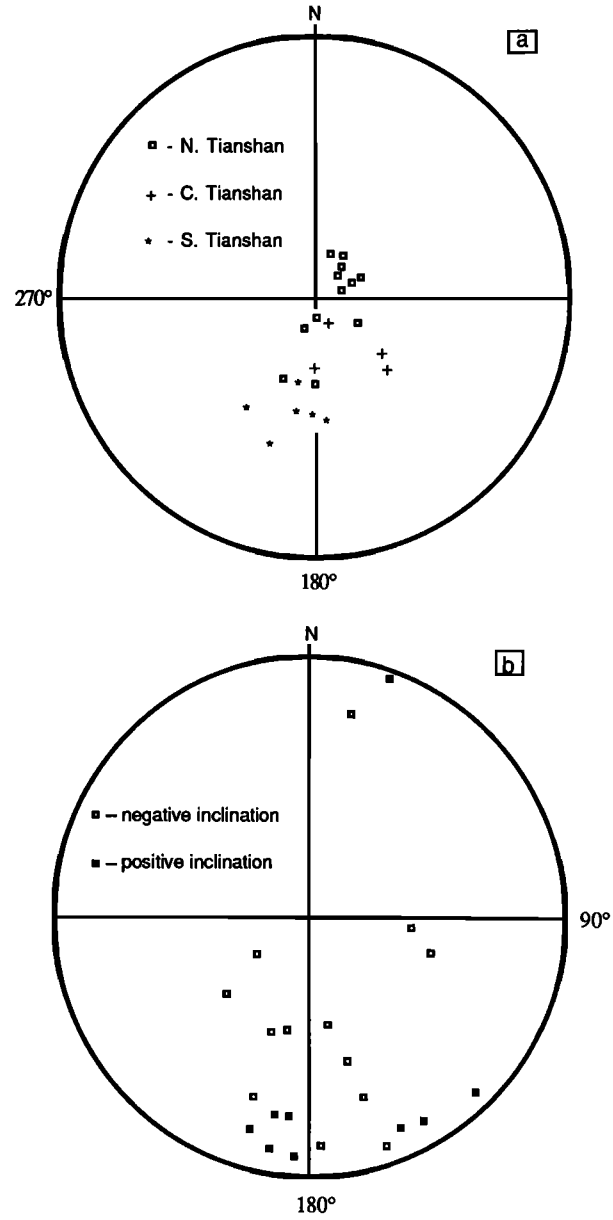


Fig. 7. (a) in-Situ and (b) tilt-corrected site mean directions from the Dushanzhi-Kuqa highway. In Figure 7a, all inclinations are negative. 22 sites from different parts of Tianshan are indicated. Note the greater inclinations and larger declination variations for northern and central Tianshan sites. Figure 7b clearly indicates a negative fold test as discussed in text. Details are listed in Table 2.

Shaaermin Formation that includes yellow muddy limestone along the north slope of southern Tianshan and dark-colored limestone at the southern peak of Tianshan.

Demagnetization and Paleomagnetic Directions

In general, a low temperature component is removed below 300°C that is clearly a recent overprint. A high temperature component can be defined with blocking temperatures of 590°C or lower for samples from 14 sites and 680°C or lower for samples from the other eight sites. All directions are listed in Table 2, while the site mean directions are graphically displayed in Figure 7 in both in situ and tilt corrected coordinates. We also sampled two sites (15 samples) from the Jurassic coal-bearing mudstones in central Tianshan, but they show only the present-day geomagnetic field.

There are three features worth noting about the paleomagnetic results from these Carboniferous and Devonian rocks across Tianshan. (1) all four formation means fail a fold test, as indicated by greater k and smaller $\alpha 95$ values in the in situ coordinates (Table 2). Simply stepwise unfolding (i.e.,

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TABLE 2. Paleomagnetic Directions From Dushanzhi-Kuqa Highway Across Tianshan

Site	Strike	Dip	n (n')	Db	Ib	Da	Ia	K	α_{95}	Range (°C)
Qiaxia Formation (C_2), Northern Tianshan										
1	161	74S	3 (3)	202	-53	106	-37	130	10.9	300-580
2	276	34N	6 (6)	200	-76	191	-42	95	6.9	550-630
4	275	85N	5 (5)	173	-81	183	4	682	2.9	300-600
5	142	60S	4 (4)	180	-53	96	-46	86	9.9	300-610
27	286	119N	7 (8)	20	-70	195	9	31	11.0	300-550
30	245	105N	8 (9)	34	-68	136	3	63	7.0	100-570
mean (N = 6)				178	-81			10	22.3	
						155	-23	3	45.3	
Shirenzhigou Formation (C_3), Northern Tianshan										
9	96	86S	5 (7)	76	-75	20	1	51	10.8	300-600
10	84	72S	4 (4)	69	-72	12	-13	601	3.7	150-610
31	255	105N*	10 (11)	42	-72	150	5	68	5.9	300-570
32	340	60N	3 (6)	119	-68	227	-43	106	12.0	100-600
33	76	72S	6 (6)	66	-71	185	14	44	10.2	300-570
34	82	68S	9 (9)	49	-76	184	14	125	4.6	100-580
mean (N = 6)				72	-74			85	7.3	
						188	1	7	28.2	
Dahajunshan Formation and Yiayungou Formation (C_1), Central Tianshan										
14	231	16N	10 (11)	181	-57	170	-44	237	3.1	300-680
16	296	65N	5 (6)	157	-79	197	-18	57	10.5	400-670
17	280	71N	5 (5)	133	-48	156	6	50	11.0	300-580
35	290	71N	5 (5)	128	-50	162	-4	41	12.1	300-550
mean (N = 4)				146	-60			18	22.4	
						171	-15	9	33.4	
Shaaermin Formation (D_2), Southern Tianshan										
19	94	32S	5 (6)	211	-36	235	-62	163	6.0	300-550
20	101	12S	9 (9)	197	-28	198	-40	444	2.4	300-550
37	287	46N	9 (10)	187	-41	189	5	59	6.8	300-550
39	206	41N	7 (10)	183	-41	163	-18	34	10.5	300-550
40	245	49N	7 (8)	191	-51	177	-7	149	5.0	300-550
41	200	21N	9 (10)	178	-40	165	-30	108	5.0	300-575
mean (N = 6)				192	-40			47	9.9	
						183	-27	7	27.1	
Overall mean (N = 22)				166	-72			8	11.7	
						174	-14	5	15.9	

Headings are n(n'), number of samples used (collected); Db, Ib (Da, Ia), declination and inclination before (after) tilt correction. C, Carboniferous; D, Devonian; 1, 2, and 3 are early, middle, and late. N is number of sites to calculate mean. * Angle > 90° indicates overturned bed.

taking present strike as the rotation axis for the beds) performed for all sites and also for each individual formation gives the best statistics in the in situ coordinates, thereby excluding the possibility of the magnetization being synfolding; (2) the in situ inclinations are all negative. Note though that their values range from over 70° to 60° in the north and central Tianshan to about 40° in the south; (3) for all sites, the in situ declinations exhibit considerable scattering, with the largest represented by those from northern Tianshan, ranging from 20° to 202° (Table 2).

In summary, the characteristic directions defined by the high-temperature components from these Carboniferous and Devonian rocks have clearly resulted from pervasive postfolding remagnetization event(s). There is a notable difference in the in situ inclinations (Table 2) between the Devonian carbonates in the southern Tianshan (mean inclination = -40°) and the Carboniferous clastics and volcanics in central and northern Tianshan (mean inclination = -65). The exact cause for this is not known. One possibility is that these rocks were remagnetized at different times. More specifically, we could argue that the higher-inclination remagnetizations of central and northern Tianshan were acquired sometime in the Permian, because the inclinations were all negative, corresponding to the Kiaman Reversed Interval, and the average mean inclination of -65° is comparable to that of the primary Permian directions found on both sides of Tianshan (see Bai et al. [1987], Li et al.[1988a], and McFadden et al. [1988] for northwestern Tarim; and this study for southern Junggar). The declinations seen in the Carboniferous rocks in northern Tianshan show significant scattering, from 20° to 202°. This may be interpreted as having resulted from subsequent tectonic rotations, but the details of these rotations cannot yet be resolved. The remagnetization directions of the Devonian carbonates in southern Tianshan, on the other hand, might

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TABLE 3. Late Permian Paleomagnetic Poles for South Junggar, Tarim, and Adjacent Blocks

Plate	Plat. / Plong.	N(n)	K	A95	No.	Reference
South Junggar	76.0 / 193.0	7(63)	24	12.6	1	this study
North Tarim	73.2 / 191.0	10(62)	45	7.3	2	McFadden et al. [1988]
North Tarim	65.6 / 181.2	21(127)	61	3.9	3	Li Y.P. et al. [1988]
South Tarim	61.1 / 181.0	7(35)	32	9.3	4	Li Y.A. et al. [1989]
Siberia	52.0 / 145.0	7	-	8.0	5	Khramov et al. [1981]
Kazakhstan	54.0 / 169.0	1	-	--	6	Khramov et al. [1981]
Russia	46.0 / 167.0	23	-	2.5	7	Khramov et al. [1981]
North China	46.1 / 0.8	10	46	6.3	8	Nie et al. [1990]
South China	42.6 / 230.2	7	18	14.8	9	Nie et al. [1990]

Plat. / Pong. are pole latitude / longitude. N (n) is number of sites or studies (samples). No. is pole number in Figure 8.

have been acquired at a later time, because the in situ directions are similar to the directions reported from the Jurassic and Cretaceous rocks by Li et al. (I/D= 202°/-40°; their Baicheng result, 1988b) from northern Tarim. We should note though, that this Mesozoic result yields paleolatitude of 25°N that appears significantly lower than what might be expected from plate reconstructions [Lin and Fuller, 1990] and the temperate Jurassic flora found in the Junggar basin [Yao, 1991]. An alternative interpretation to our Devonian remagnetization directions could be that they were also acquired sometime in the Permian, but subsequent block tilting has caused the lowering of the inclinations.

DISCUSSIONS AND CONCLUSIONS

We interpret the characteristic direction from the Heavenly Lake section to represent a primary Late Permian magnetization. The corresponding paleomagnetic pole is at 76°N, 193°E, labeled pole 1 in Table 3 and Figure 8. Two Late Permian poles from northern Tarim (pole 2, [McFadden et al., 1988], and pole 3, [Li et al., 1988a]) and one from Hotan of southwestern Tarim (pole 4, [Li et al., 1989]) are also included in Table 3 and Figure 8, along with five other Late Permian poles from adjacent Eurasian continental plates. The result from Li et al. [1988a] is from the Permian basaltic dykes in northwestern Tarim and passes a fold test, whereas the Late Permian and Early Triassic directions of McFadden et al. [1988] were obtained from a sedimentary sequence north of Kuqa, which is characterized by a repetition of normal and reversed polarities. The results from southwestern Tarim [Li et al., 1989] are derived from a red terrestrial clastic unit and include only reversed polarity. These four Late Permian poles (Figure 8, poles 1 to 4) are statistically indistinguishable, with overlapping confidence circles. Therefore the first conclusion is that by the Late Permian there was no significant separation between Tarim and Junggar, then the southern margin of the already amalgamate Kazakhstan and Siberia, supporting an Early Permian closure of the Tianshan suture.

We have also shown that the Devonian and Carboniferous rocks across Tianshan were all remagnetized along the Dushanzhi-Kuqa highway. In particular we interpret the remagnetization direction of the Carboniferous clastics and volcanics from northern and central Tianshan to be Permian in age. Interestingly, reversed postfolding directions of Permian age have also been reported from several other late Paleozoic sequences around Tianshan, including the Lower Permian sedimentary rocks in northwestern Tarim [Sharps et al., 1989] and the Carboniferous to Permian igneous rocks in western Junggar [Zhao and Coe, 1990]. This generally southwesterly and steeply upward direction probably reflects a major collision-induced remagnetization event (thermal or chemical) in the Permian.

Finally, when one compares the Late Permian

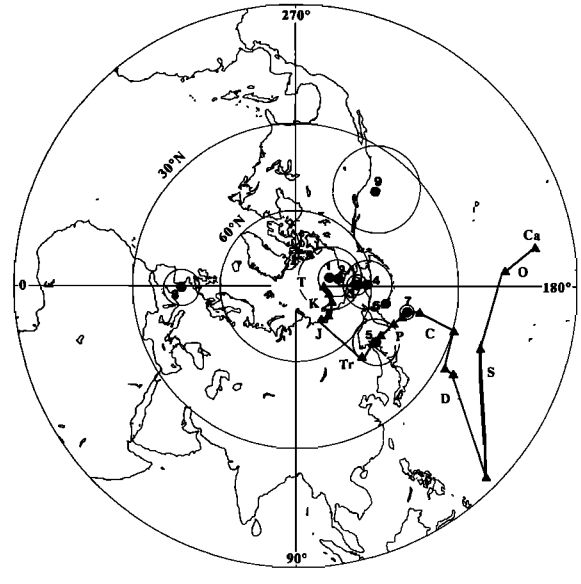


Fig. 8. Polar projection of Late Permian paleomagnetic poles from Eurasian plates and Phanerozoic apparent polar wonder path for stable Europe. Numbers represent paleomagnetic poles for different plates: 1, southern Junggar (this study); 2, northwestern Tarim [McFadden et al. 1988]; 3, northwestern Tarim [Li et al. 1988a]; 4, southern Tarim [Li et al. 1989]; 5, Siberia; 6, Kazakhstan; and 7, Russia [Khramov et al. 1981]; 8, North China; and 9, South China (from summary by Nie et al. 1990); Apparent polar wonder path for stable Europe is based on data from Van der Voo [1990]. Geological periods for different segments on the APW path are indicated: Ca, Cambrian; O, Ordovician; S, Silurian; D, Devonian; C, Carboniferous; P, Permian; Tr, Triassic; J, Jurassic; K, Cretaceous; and T, Tertiary. All data are listed in Tables 3 and 4.

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paleomagnetic poles from the major Eurasian tectonic elements (Figure 8 and Table 3), it is evident that the Late Permian poles from Tarim and the Heavenly Lake section are markedly different from those of North China (pole 8) and South China (pole 9), implying that these were clearly separate tectonic elements at that time. The poles from northwestern China are also notably different from those from the Soviet Union [Khramov et al., 1981], in particular that of Siberia (pole 5) and Russia (pole 7), and to a lesser extent, that of Kazakhstan (pole 6). This difference is not to be expected if Tarim had become a part of Eurasia by the Late Permian as has been concluded earlier, which is why McFadden et al. [1988] proposed the existence of a wide ocean between Tarim and Kazakhstan at this time. It should be noted that the paleolatitude of 39°N for the Heavenly Lake (44°N, 88.1°E) using our paleomagnetic pole is identical to that calculated from the Kazakhstan Late Permian pole. Similar conclusions can also be drawn by using the results from northwestern Tarim and Siberia or Russia. In other words, the differences in these Late Permian paleomagnetic poles are primarily the result of the differences in the declinations, which can be the result of a number of possibilities. First, one could argue that the paleomagnetic results from the Soviet Union might have serious flaws [e.g., Lee et al., 1987]. But if we use the apparent polar wonder path of stable Europe (Figure 8 and Table 4, data from Van der Voo [1990]) to represent the whole of Eurasia, it is clear that the Late Permian poles from at least Siberia (pole 5) and Russia (pole 7) fall very close to the Permian part of the APW path (Figure 8). It is therefore not unreasonable to believe that the Late Permian poles from the Soviet Union represent relatively reliable directions. The second possibility is that the Late Permian poles from northwestern China are secondary rather than primary, as they do seem to fall closer to the Late Cretaceous and Tertiary rather than the Permian part of the APW path of stable Europe (Figure 8). Although this remains possible, the similarity alone does not mean that these Late Permian poles from Tarim and southern Junggar have to be a Cretaceous or Tertiary overprint. In fact, field tests for these Late Permian results are highly significant. In particular, the repetition of dual polarity directions reported by McFadden et al. [1988] strongly suggests that a simple Mesozoic and Cenozoic remagnetization is not likely. The third possibility is that subsequent relative rotations between Tarim/southern Junggar and Eurasia have given rise to the differences in the declinations and therefore pole positions. Furthermore, the rotations have to be regional instead of local in nature since they have similarly affected both southern Junggar and Tarim.

To understand this better, we need to address a somewhat larger problem of Permian Asian plate reconstruction. Figure 9 is a Late Permian global reconstruction, modified after Nie et al. [1990], which emphasizes the Asian elements. The positions of North China, South China, and Tarim are primarily based on paleomagnetic results (Table 3), with additional constraints from geological, paleoclimatic and biogeographic data [Ziegler, 1990; Nie, 1991]. The locations of other microcontinents, such as Lhasa, Western Qiangtang, Eastern Qiangtang, and Indochina, can be determined only to the extent of either being marginal to Gondwana or within the Paleo-Tethys. Traditionally, the so-called Central Asian Foldbelt, or the Altaiides, including the vast area between Siberia in the north and Tarim and North China in the south, has been considered to be a wide and complex boundary zone, marked by numerous sutures separating a collage of arc-related and continental fragments that all closed prior to the late Paleozoic [e.g., Yang et al., 1986; Wang and Liu, 1986]. Based on this view, it has been hard to understand why the Late Permian and early Mesozoic paleomagnetic directions from China and Siberia are so different [Lin et al., 1985; Opdyke et al., 1986; Zhao and Coe, 1987]. We [Rowley et al., 1985; Nie et al., 1990], following Kosygin and Parfenov (1981) and Parfenov and Natal'in (1985), point out that the Mongolo-Okhotsk suture (= the Dzhagdi suture) is a key to the understanding of the late Paleozoic to Mesozoic tectonic history between Siberia, Mongolia, and North China. This suture is presently situated within the greater Altaiides to the north of China along the southeastern margin of Siberia and extends westward into central Mongolia (Figure 1). It had a diachronous closing history, as inferred from termination of marine sedimentation, development of fold belts and unconformities, and also distribution patterns of arc-related magmatism. Earliest closure occurred in the west in central Mongolia in the Permian, while in the east near the Sea of Okhotsk it remained as an open ocean until Late Jurassic to Early Cretaceous times (for a more updated review, see Zonanshain et al. [1990]). Therefore we believe that the so-called "Hercynian" orogenic event recorded along the Central Asian Foldbelt, was the result of the collision between North China and the Mongolian arc, not Siberia sensu stricto [Nie et al., 1990]. The apparent termination of the Mongolo-Okhostk suture in central Mongolia suggests that the Mongolian arc was attached to Siberia at its western end in the Permian. The Mongolian arc, along with the Dahinggan and Xingkai-Bureya blocks in northeast China (Figure 1), was probably part of Siberia for much of the Paleozoic, judging from the similarity of the Cambrian faunas [Yin, 1989], the occurrences of the Silurian brachiopod *Tuvaella* [Su, 1981; Rong and Zhang, 1982], and dominance of the late Paleozoic Angaran flora [Li and Yao, 1983]. (We note that our interpretation is different from that of Zonanshain et al. [1990], who argued that southern Mongolia and northeastern China were separated from Siberia at least in the Devonian, in part because of the apparent inconsistency between the Devonian reefal carbonates found south of the Mongolo-Okhotsk suture and the high paleolatitude (> 60°N) of Siberia inferred from Siberia's paleomagnetic data. How to reconcile the paleomagnetic results with the Devonian carbonates and marine evaporitic sequences of not only southern Mongolia but also Siberia [Ronov et al., 1984] remains to be understood.) The partial separation of the Mongolian arc from Siberia prior to the Permian may have had a geometry similar to the Sea of Japan

TABLE 4. Paleomagnetic Poles for Stable Eurasia

Age	Plat. / Plong.	N	K	A95
T1	78 / 178	19	77	3.8
K2	74 / 155	3	86	3.4
K1	76 / 167	2	-	--
J2 - J3	72 / 137	7	60	7.9
J1 - J2	73 / 129	3	85	13.4
Tr2 - Tr3	52 / 133	5	29	14.4
Tr1 - Tr2	52 / 150	13	103	4.1
P2	49 / 159	5	69	9.3
P1	47 / 164	20	128	2.9
C3 - P1	41 / 168	17	184	2.6
D3 - C2	29 / 164	3	46	18.4
D2 - D3	27 / 151	6	38	10.9
D1 - D2	24 / 151	5	82	8.5
S3 - D1	2 / 135	13	23	8.9
S2 - S3	20 / 161	11	44	6.9
O2 - S1	16 / 184	1	-	--
Ca1	8 / 189	1	-	--

Poles are from Van der Voo [1990]. Plat. / Plong. are pole latitude and longitude; N is number of independent studies; K is precision parameter; A95 is radius of 95% confidence cone. Ages are T, Tertiary; K, Cretaceous; J, Jurassic; Tr, Triassic; P, Permian; C, Carboniferous; D, Devonian; S, Silurian; O, Ordovician; and Ca, Cambrian. The subscripts 1, 2, and 3 refer to early, middle, and late series.

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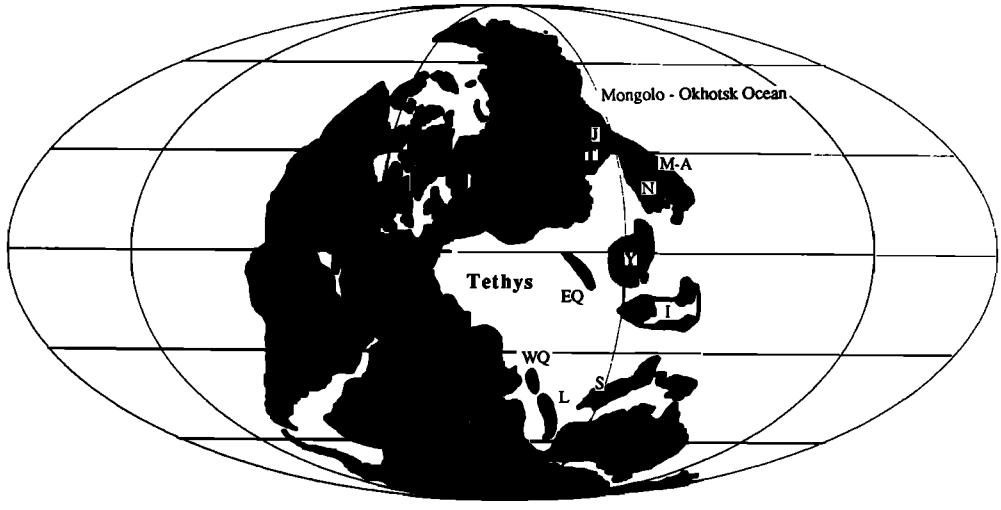


Fig. 9. Late Permian global reconstruction (modified after Nie et al. [1990]). Paleomagnetic results are used to reconstruct Tarim (T), Junggar (J), South China (Y), and North China (N) in addition to other geological constraints. Phytogeographic and paleoclimatic data [Ziegler, 1990; Nie, 1991] are used to reconstruct approximate positions of Eastern Qiangtang (EQ), Indochina (I), Lhasa (L), Western Qiangtang, and Sibumasu (S) blocks. Notice the positions of the Mongolian arc (M-A) and the Mongolo-Okhotsk ocean between Eurasia and North China.

as indicated in Figure 9. Collision of the Mongolian arc with North China in the Late Permian, coupled with clockwise rotation of Siberia is responsible for the final closure of this ocean in the Late Jurassic to Early Cretaceous, which in turn can be used to explain the discordance of the Permian and early Mesozoic paleomagnetic poles.

The fact that the Late Permian poles from northwestern China are also different from that of Siberia suggests that the effect of the closure of this suture and associated relative motion may have extended farther to the west than the termination of this suture in the Khangay highland in central Mongolia. In fact, the sense of clockwise rotation to bring the Late Permian poles from northwestern China (poles 1-4, Figure 8) to match those from the Soviet Union (poles 5-7) is compatible with the sense of rotation to reopen the Mongolo-Okhotsk ocean (Figures 1 and 9). As there is no evidence for a late Paleozoic or early Mesozoic ocean westward beyond central Mongolia, the relative motion between Tarim/southern Junggar and Siberia must have been distributed entirely within the continental realm to the north of northern Tianshan. The presence of the Triassic and Jurassic strike-slip faults and associated basins within Mongolia [Lefeld, 1978] could have resulted from these tectonic rotations. In other words, we believe that although Tarim and southern Junggar were physically attached to Eurasia as early as in the Late Permian as we show in our reconstruction (Figure 9), there have been significant subsequent relative motions. It also follows that even though southern Junggar and northern Tianshan were tectonically part of Eurasia long before the Permian, they appear to have reacted to later tectonic motion in a way more similar to that of Tarim, rather than Kazakhstan or Siberia, after the collision along Tianshan. This disjoint nature of tectonism is probably not atypical for the part of a continental plate that is involved in the foreland folding and thrusting.

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